Excitation-Preserving Distributed Safety Filter for Multi-Agent Adaptive Control on Complex Dubins Vehicles

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***Abstract*—For a network of N complex-Dubins agents with unknown loss-of-effectiveness (LOE) on the turn-rate channel λ_i, we identify the convergence rate of the parameter estimate $\hat{\theta}_i \to 1/\lambda_i$ under a distributed higher-order CBF (HOCBF) safety filter as the constrained scalar Fisher information of $1/\lambda_i$ on the binary freedom cone $F_i \in \{\{0\}, \mathbb{R}\}$—equivalently, the inverse of the corresponding scalar Cramér–Rao bound. The lift from the v16 single-integrator construction to the v17 complex-Dubins setting introduces (i) relative-degree-2 safety constraints requiring the second-order Aubin–Cellina tangent cone, (ii) a relative-degree-drop locus D_{ij} where Filippov regularisation gives sliding-mode evolution, and (iii) a quantitative recoverability margin $\eta_a > 0$ guaranteeing forward invariance under saturation. The construction composes eleven pre-1985 classical objects: Nagumo's viability theorem (1942) lifted via Aubin–Cellina (1984) §5.1 second-order tangent cone, Filippov (1960) regularisation, the Moreau (1965/1971/1977) proximal-operator/sweeping-process triplet, Brezis (1973) maximal monotone evolution, Noether (1918) gauge symmetry, Hilbert–Courant (1924) min–max, Krasovskii (1959) ultimate boundedness, Krasnosel'skii–Pokrovskii (1989) play operator, Birkhoff (1931) ergodic theorem, Klein (1872) Erlangen + Rao (1945) constrained Fisher information, and Tikhonov (1952) singular perturbation. The OSQP scalar QP at $h_{\text{outer}} = 5\,\text{ms}$ is real-time feasible up to $N \approx 4$ on a desktop CPU and at $h_{\text{outer}} = 10\,\text{ms}$ up to $N = 8$ ($p_{99} = 3.89\,\text{ms}$ serial, 38.9% utilisation). Communication-delay robustness improves $20\times$ over v16 ($5\,\text{ms} \to 100\,\text{ms}$ with $h_{ij} \geq 0$ preserved), verified at both $N = 4$ (cross-swap) and $N = 8$ (rotating-ring rosette), due to the analytical R3 latency residual in the CBF tightening $\delta_{ij}(t)$.**

I. Problem statement

A team of N complex-Dubins agents must reach a target formation $\{t_i\}_{i=1}^N \subset \mathbb{C}$ while (i) maintaining pairwise safety $|r_i - r_j| \geq r_{\text{safe}}$, (ii) identifying each agent's unknown LOE gain $\lambda_i \in [\lambda_{\min}, 1]$ on the turn-rate channel, and (iii) using only neighbour-broadcast information $\{r_j, v_{a,j}, \hat{\theta}_j, u_{2,j}^{\text{safe}}(t^-)\}$ from a connected communication graph $\mathcal{G}$. We extend the single-agent adaptive-Dubins formulation of Maldonado-Naranjo–Annaswamy (2025) [1] to the multi-agent setting with HOCBF safety constraints (Xiao–Belta 2021) [2] and prove the convergence rate of $\hat{\theta}_i$ as the constrained Cramér–Rao bound on the binary freedom cone (Rao 1945 [3], Whitney 1965 stratification [4], Amari 1985 information geometry [5]).

II. Construction

A. Plant, reference, adaptive law, Kalman–Bucy auxiliary

For agent $i \in \{1, \ldots, N\}$ with state $(r_i, v_{a,i}) \in \mathbb{C}^2$, position $r_i = x_i + iy_i$, velocity-along-heading $v_{a,i} = V_0 e^{i\psi_i}$ at constant speed V_0:

$$\dot{r}_i = v_{a,i}, \qquad \dot{v}_{a,i} = i\,\lambda_i\,u_{2,i}\,v_{a,i},$$

with scalar real turn-rate $u_{2,i} \in \mathbb{R}$. By Carathéodory (1909) accessibility [6] and Chow (1939) bracket theorem [7], the system is reachable on $SE(2)$ via $u_{2,i}$ alone—the Lie bracket of the forward-motion vector field $X_1 = v_{a,i}\partial_{r_i}$ and the turning vector field $X_2 = iv_{a,i}u_{2,i}\partial_{v_{a,i}}$ generates the missing lateral direction.

The reference, adaptive law, and KB filter take the complex-regressor form (paper §2):

$$\begin{aligned}
u_{2,i}^{\text{ref}} &= u_{2,\text{ref},i} - \Re\Big(\frac{K_T e_i + K_F \sum_j (e_i - e_j) - \cdots]}{i\,v_{a,i}}\Big),\\
u_{2,i}^{\text{AC}} &= \hat{\theta}_i\,u_{2,i}^{\text{ref}}, \quad \tilde{v}_i := v_{a,i} - v_{\text{ref},i},\\
\dot{\hat{\theta}}_i &= \text{Proj}_{[\theta_{\min},\theta_{\max}]}\Big[-\tfrac{\gamma}{m_i^2}\,\Re\big(\overline{iv_{a,i}u_{2,i}^{\text{ref}}}\,\tilde{v}_i\big)\Big],
\end{aligned}$$

with $m_i^2 := 1 + V_0^2(u_{2,i}^{\text{ref}})^2$, the Pomet–Praly normalisation (Pomet–Praly 1992) [8]. The KB filter $\dot{P}_i = -P_i^2|v_{a,i}u_{2,i}^{\text{ref}}|^2/m_i^2$ runs in parallel; under closed-loop PE, $P_i \to 0$ exponentially (Anderson 1985 [9]; modern Robbins–Monro / Borkar 2008 framework [10], [11]).

B. HOCBF + Filippov regularisation at the locus

The pairwise CBF $h_{ij}(r) := |r_i - r_j|^2 - r_{\text{safe}}^2$ has **relative degree 2** w.r.t. u_2 (Aubin–Cellina 1984 §5.1 second-order tangent cone [12]; Xiao–Belta 2021 HOCBF [2]). Define $\psi_{1,ij} := \dot{h}_{ij} + \alpha_1 h_{ij}$; the HOCBF condition $\dot{\psi}_{1,ij} + \alpha_2\psi_{1,ij} \geq 0$ multiplied through by $\hat{\theta}_i$ yields the gauge-fixed constraint

$$\begin{aligned}
c_{ij}(u_{2,i}; r, v_a, \hat{\theta}) := {}& 2\,u_{2,i}\,\Im\big((r_i - r_j)\overline{v_{a,i}}\big)\\
&- 2\tfrac{\hat{\theta}_i}{\hat{\theta}_j}u_{2,j}^{\text{safe}}(t^-)\,\Im\big((r_i - r_j)\overline{v_{a,j}}\big)\\
&+ \hat{\theta}_i\,b_{ij}^{(0)} \;\geq\; \delta_{ij}(t),
\end{aligned}$$

with $b_{ij}^{(0)} = 2|v_{a,i} - v_{a,j}|^2 + 2(\alpha_1 + \alpha_2)\Re\big((r_i - r_j)\overline{v_{a,i} - v_{a,j}}\big) + \alpha_1\alpha_2 h_{ij}$ the corrected $\hat{\theta}$-independent residual.

Relative-degree-drop locus. At $\arg(r_i - r_j) - \arg v_{a,i} \in \{0, \pi\}$, the HOCBF coefficient $a_{ii} = 2\Im\big((r_i - r_j)\overline{v_{a,i}}\big)$ vanishes; the differential inclusion becomes non-Lipschitz. We adopt **Filippov (1960)** regularisation [13]: the convex-hull set-valued vector field gives an absolutely-continuous solution with sliding-mode behaviour along D_{ij}. The slack-penalty QP

is the smoothed Tikhonov approximation (Tikhonov 1963 / Engl–Hanke–Neubauer 1996 [14], [15]) with $\mathcal{O}(M^{-1/2})$ rate.

Recoverability margin (paper §3.1.1). Axiom (A3") requires $|a_{ii}| \geq \eta_a > 0$ on the closed-loop invariant set, with η_a chosen so $\dot{\psi}_{\max} \cdot \eta_a \geq |\text{res}_{ij}|_{\text{worst}}$. For the §8.3 parameters, $\eta_a^{\text{practical}} \approx 1.6$.

$\delta_{ij}(t)$ **residual aggregation (paper §3.1.2).** Three sources—(R1) gauge-fix LHS-coefficient residual, (R2) cross-term substitution residual, (R3) broadcast-latency residual—aggregate into

$$\delta_{ij}(t) = \dot{\psi}_{\max}[|a_{ii}|\varepsilon_i^{(1)} + |a_{ij}|\varepsilon_{ij}^{(2)}] + \tfrac{\hat{\theta}_i}{\hat{\theta}_j}|a_{ij}|\varepsilon_j^{(3)}(\tau_d) + \zeta_0,$$

where $\varepsilon_j^{(3)}(\tau_d) = L_{\text{QP}} \cdot \tau_d \approx 0.0094$ at $\tau_d = 5\,\text{ms}$—the analytical absorbing of broadcast latency that yields the $20\times$ communication-delay-robustness improvement over v16.

C. QP-resolvent (Moreau prox + KKT closed form)

The closed-loop control is the Moreau proximal operator (Moreau 1965/1971/1977 [16]–[18]) on the time-varying interval $K_i(t) \subset \mathbb{R}$, satisfying the prox-regularity hypothesis (H-PR) (Edmond–Thibault 2006 [19]). The safety filter is a Maxwell-governor (Maxwell 1868 [20]) on the AC command:

$$u_{2,i}^{\text{safe}} = \arg\min_{u_2, s \geq 0} (u_2 - (u_{2,i}^{\text{AC}} + \tilde{e}_i^{\text{pe}}))^2 + M \sum_{j \in \mathcal{N}_i^{\text{on}}} s_{ij}^2$$
$$\text{s.t. } c_{ij}(u_2; \cdots) + s_{ij} \geq \delta_{ij}(t),\ s_{ij} \geq 0,\ |u_2| \leq \dot{\psi}_{\max}. \quad (1)$$

For $|\mathcal{N}_i^{\text{on}}| = 1$, the closed-form Karush–Kuhn–Tucker (KKT, Karush 1939 [21] / Kuhn–Tucker 1951 [22]) solution is $u_{2,i}^{\text{safe}} = (u_{2,i}^{\text{AC}} + \tilde{e}_i^{\text{pe}}) + \mu_{ij}^* a_{ii}$ with $\mu_{ij}^* = \max(0, (\delta_{ij} - c_{ij})/a_{ii}^2)$, satisfying complementary slackness $\mu_{ij}^*(c_{ij} - \delta_{ij}) = 0$. For $|\mathcal{N}_i^{\text{on}}| \geq 2$ (multi-pair active set), no closed-form Lagrangian exists; OSQP solves the QP numerically. The exact-penalty threshold is $M^* > \|\lambda^*\|_\infty$ (Bertsekas 1999 §5.4 [23]); empirically M scales as $\sqrt{N(N-1)/2}$ with the active-set cardinality.

Empirical OSQP timing (Boyd [24], Passes 31/37). At $N = 4$ (6 pairs): $0.31\,\text{ms}$ warm / $5.3\,\text{ms}$ cold. At $N = 8$ (28 pairs, $\text{K}_8 antipodal - ringrosette, worst - caseall7neighboursactiveforoneagent$) :$\text{p}_{50} = 0.343\,\text{ms}$, $p_{99} = 0.486\,\text{ms}$ warm-started; $2.74\,\text{ms}$ cold-start. Serial 8-agent p_{99}: $3.89\,\text{ms}$ / $h_{\text{outer}} = 10\,\text{ms} = 38.9\%$ utilisation (well within budget). The §VI N=8 demo bumps h_{outer} $5 \to 10\,\text{ms}$ (Tikhonov margin $\varepsilon = K_T \Lambda_{\min} h_{\text{outer}} = 0.024$, separation factor $42\times$); §VI's $N = 2$ proof-bearing example keeps the baseline $5\,\text{ms}$.

III. Symmetry, Lyapunov, identifiability

A. Gauge $\mathbb{R}_{>0} \times U(1)$ + shape-space reduction

The Lyapunov function $V = \sum_i V_i + V_{\text{form}} + \kappa \sum_{i<j} B(h_{ij})$ with $V_i = (|e_i|^2 + |\tilde{v}_i|^2)/(2m_i^2) + (\lambda_i/2\gamma)\tilde{\theta}_i^2$ (Noether-required $\kappa_v = 1$, Lie 1880s [25]) is invariant under the Klein-Erlangen gauge $\mathbb{R}_{>0}$ on $\hat{\theta}$ and the Weyl gauge $U(1)$ on heading rotation (Klein 1872 [26], Weyl 1918 [27], Noether 1918 [28]). The cross-swap (§VI) is a **rotational relative equilibrium** (Leonard et al. 2007 [29]) of the reduced shape-space dynamics on $SE(2)^N/U(1)$.

B. Lemma 2 (Krasovskii UUB, v17 form)

$\dot{V} \leq -\eta(\sum_i(|e_i|^2 + |\tilde{v}_i|^2)/m_i^4 + \|\xi\|^2) + \mathcal{O}(A_e^2 + \sup_i P_i^2 + M^{-1/2})$, with $\eta \geq \min(K_T/2, \lambda_{\min}/2 - \bar{\mu}^2 L_{\text{QP}}^{*2}/(2K_T))$ via Pomet–Praly cancellation (Noether-conserved current of G_κ), Hilbert–Courant min-max (1924) [30] on the **Kirchhoff Laplacian** $L_{\mathcal{G}}$ (Kirchhoff 1847 [31], Fiedler 1973 algebraic connectivity [32]), and Tikhonov (1952) singular perturbation [33] for the heading-PD inner-loop cascade (Maldonado-Naranjo–Annaswamy 2025 §III.B [1]). At §8.3 parameters: $K_T \lambda_{\min} = 2.4$ vs threshold $\bar{\mu}^2 L_{\text{QP}}^{*2} \approx 0.32$—margin $7.5\times$.

C. Birkhoff identifiability + scalar Fisher info

The binary freedom cone $F_i \in \{\{0\}, \mathbb{R}\}$ (cruising vs. active-pair) Whitney-stratifies the closed-loop measure μ (Whitney 1965 [4], Krasnosel'skii–Pokrovskii 1989 hysteresis [34], Davis 1984 PDMP [35]). The constrained scalar Fisher information of $1/\lambda_i$ is

$$\boxed{\bar{\rho}_i := \mathbb{E}_\mu\Big[\tfrac{V_0^2(u_{2,i}^{\text{ref}})^2}{1+V_0^2(u_{2,i}^{\text{ref}})^2} \cdot \nVdash\{\mathcal{N}_i^{\text{on}} = \emptyset\}\Big] \in [0,1).} \quad (2)$$

This is the Amari (1985) mixture-coordinate framing on the Whitney-stratified parameter manifold (Fisher 1925 [36] + Rao 1945 §6 [3] + Stoica–Ng 1998 engineering CRB [37]). The convergence rate is $\rho_i = \gamma\bar{\rho}_i$.

IV. Theorem

[Excitation-preserving distributed safety filter, complex Dubins, v17.1] Under axioms (A1) projected $\hat{\theta}_i \in [\theta_{\min}, \theta_{\max}]$, (A2)+(A2') closed-loop PE, (A2") free-time dwell, (A3") HOCBF initial conditions $h_{ij}(0) \geq \delta_{ij}(0) + \zeta$, $\psi_{1,ij}(0) \geq 0$, $|a_{ii}| \geq \eta_a > 0$, (A4) continuous broadcast of $\{r_j, v_{a,j}, \hat{\theta}_j, u_{2,j}^{\text{safe}}(t^-)\}$, and (A5) active-set non-saturation ($|\mathcal{N}_i^{\text{on}}| \leq 1$), the closed-loop trajectories of $\dot{r}_i = v_{a,i}$, $\dot{v}_{a,i} = i\lambda_i u_{2,i}^{\text{safe}} v_{a,i}$, $|v_{a,i}| = V_0$ generated by Moreau's (1971) sweeping process off the locus D_{ij} and Filippov's (1960) regularisation on D_{ij} satisfy:

(1) $h_{ij}(r(t)) \geq 0$ for all $t \geq 0$, by Nagumo's viability theorem (1942) [38] lifted via Aubin–Cellina (1984) §5.1 second-order tangent cone [12], with sliding-mode behaviour on D_{ij};
(2) ultimate boundedness $V(t) \leq V(0)e^{-\eta t} + \mathcal{O}(A_e^2/\eta) + \mathcal{O}(\sup_i P_i^2/\eta) + \mathcal{O}(M^{-1/2}/\eta)$ [39];
(3) scalar parameter convergence $\hat{\theta}_i \to 1/\lambda_i$ exponentially with rate $\rho_i = \gamma\bar{\rho}_i$, where $\bar{\rho}_i$ is the constrained scalar Fisher information (2).

V. Proof outline (six lemmas, all classical)

1) **Lemma 5.1 (forward invariance).** Aubin–Cellina + Filippov + Krasovskii reverse-Lyapunov on $\psi_{1,ij}$. On-locus dwell-time (Pass 33 re-derivation): $\tau_d \geq \varepsilon_{\text{hyst}}/(L_{\text{eff}}|c_{ij}|'_{\max}) \gtrsim 10^{-3}$ ($\sim 1\,\text{ms}$ physical at §8.3 parameters; D_{ij} is μ-measure-zero).

2) **Lemma 5.2 (estimation tolerance).** R1+R2+R3 residual aggregation; $\delta_{ij}(t) \to \zeta_0$ as $P_i, P_j \to 0$ (Anderson 1985 [9]).
3) **Lemma 5.3 (Krasovskii UUB).** As §III.B above.
4) **Lemma 5.4 (resolvent well-posedness).** Moreau prox + Brezis 1973 [40] + Filippov on D_{ij} + Hager 1979 [41] Lipschitz with $L^*_{\text{QP}} = (1+\kappa_\lambda)/\eta_a$.
5) **Lemma 5.5 (Birkhoff identifiability).** Birkhoff 1931 [42] ergodic theorem on the K-P / PDMP invariant measure μ; $\bar{\rho}_i > 0$ under (A2')∩(A2").
6) **Lemma 5.6 (uniform multi-pair dwell-time).** K-P 1989 BV continuity + Hager 1979; $\tau_d > 0$ uniformly.

VI. Worked examples

A. $N = 2$ *head-on (relative-degree drop, slack-absorbed)*

For $r_1(0) = -L$, $r_2(0) = +L$, $v_{a,i}(0) = \pm V_0$ along the real axis: $a_{11}(0) = 0$ — the head-on configuration is on the relative-degree-drop locus D_{12}. Empirically, the QP-resolvent *deflects unconditionally* at this locus regardless of PE: the soft-slack penalty M absorbs the constraint deficit while the gauge-fixed cross-term $\hat{\theta}_i/\hat{\theta}_j$ in the HOCBF (§**??**) breaks the symmetry that would otherwise keep agents on the locus, and the recoverability margin η_a accumulates as $|a_{ii}(t)|$ grows along the deflected trajectory (Fig. 7, both panels). **Engineering implication:** the safety filter contract (forward invariance of $\{h_{ij} \geq 0\}$) is satisfied on the relative-degree-drop locus *without* PE; PE is required for the *identification* contract ($\hat{\theta}_i \to 1/\lambda_i$), not for safety. This is a strengthening of the v17.1 narrative — the paper now cleanly separates the CBF safety claim from the adaptive identification claim. Dubins shortest-path recovery (Dubins 1957 [43]; LaValle 2006 §15.3 [44]) takes $L_{\text{Dubins}} = V_0\pi/(2\dot{\psi}_{\max}) \approx 0.314\,\text{s}$; closing time-to-$r_{\text{safe}}$ is $\sim 0.8\,\text{s}$ — margin $2.5\times$. The Lemma 6 on-locus dwell-time $\tau_d \gtrsim 1\,\text{ms}$ remains the analytical guarantee, irrespective of PE; PE only changes the trajectory *after* the deflection.

B. $N = 4$ *cross-swap (proof-bearing example)*

Agents at $(\pm 3, \pm 3) \in \mathbb{C}$ swap diagonally with sinusoidal targets of period $T_{\text{swap}} = 8\,\text{s}$. Target peak velocity $|\dot{t}_i|_{\max} \approx 3.33\,\text{m/s}$ exceeds $V_0 = 1\,\text{m/s}$ by $3\times$ (Carathéodory accessibility violated for instantaneous tracking; agents follow the trend, the realistic UAV-path-planner regime). For $\bar{\mu} = 0.3$, $A_e = 0.5\,\text{rad/s}$: $\bar{\rho}_1 \approx (1-\bar{\mu}) \cdot 0.2 \approx 0.14$ giving convergence rate $\rho_1 = \gamma\bar{\rho}_1 \approx 0.021/\text{s}$.

C. $N = 8$ *antipodal-ring rosette (empirical showcase, v17.2)*

The empirical figures in §VII are computed on a regular octagon (Klein 1872 Erlangen [26]; Coxeter 1969 §2.7 [45]) with vertices $r_k(0) = Re^{i2\pi k/8}$, $R = 3\,\text{m}$, $k = 0, \ldots, 7$, performing the antipodal swap $r_k \leftrightarrow r_{k+4 \bmod 8}$ via sinusoidal targets of period $T_{\text{swap}} = 8\,\text{s}$. Communication graph is K_8 ($\binom{8}{2} = 28$ HOCBF pairs). One-way Dubins reachability (Carathéodory 1909 [6]; Frazzoli) requires $T_{\text{swap}} \geq \pi R/V_0 \approx 9.42\,\text{s}$; at $T_{\text{swap}} = 8\,\text{s}$ the peak target velocity $\pi \cdot 2R/T_{\text{swap}} \approx 2.36\,\text{m/s} > V_0$ — trend-tracking regime, as in §VI's $N = 4$ example.

The full geometric symmetry is the dihedral group $D_8 \ltimes U(1)^8$, where the action is genuinely semidirect: the 8 rotations in $D_8 \cong I_2(8)$ permute the agent indices preserving heading orientation ($\psi_k \mapsto \psi_{\sigma(k)}$), while the 8 reflections both permute indices *and* invert $U(1)$ orientation ($\psi_k \mapsto -\psi_{\sigma(k)}$); the action is thus non-trivial on the abelian fibre. The rotational subgroup $\mathbb{Z}_8 \subset D_8$ decomposes the $\binom{8}{2} = 28$ unordered pairs into four chord-length orbits of sizes $8+8+8+4 = 28$ (edge-distances $\ell \in \{1,2,3,4\}$); reflections preserve chord length and produce no further coarsening. By the Burnside / orbit-counting lemma in finite-group invariant theory (Hilbert 1890 *Über die Theorie der algebraischen Formen* [46]; cf. Weyl 1939 [47]), the formation Lyapunov factors as

$$V^{(N=8)}_{\text{form}} = \sum_{i<j} B(h_{ij}) = 8B(h^{(1)}) + 8B(h^{(2)}) + 8B(h^{(3)}) + 4B(h^{(4)}), \tag{3}$$

where $h^{(\ell)}$ is the chord-ℓ representative; the convergence-rate analysis of §III therefore reduces to four orbit representatives rather than 28 pairwise terms. (Noether 1918 [28] continues to anchor the *continuous* $\mathbb{R}_{>0} \times U(1)^N$ gauge of §III; the discrete $\mathbb{Z}_8$ reduction here is a finite-group orbit count, not a Noetherian conserved current.) The Kirchhoff Laplacian $L(K_8)$ has spectrum $\{0, 8^{(7)}\}$ (Kirchhoff 1847 [31]; Fiedler 1973 [32]); algebraic connectivity $\lambda_2(L(K_8)) = 8$, twice $\lambda_2(L(K_4)) = 4$, tightens §III's cohesion bound by $2\times$ on each orbit representative.

This is the canonical multi-agent ring benchmark of Sepulchre–Paley–Leonard 2007 [48] and Mesbahi–Egerstedt 2010 [49].

Numerical scheme overrides for this demo: $h_{\text{outer}} = 10\,\text{ms}$ (vs. $5\,\text{ms}$ baseline; Tikhonov 1952 [33] margin $\varepsilon = K_T\Lambda_{\min}h_{\text{outer}} = 0.022 \ll 1$, separation factor $42\times$. Note $K_T\Lambda_{\min}$ degrades from 2.4 at $N = 4$ to 2.2 at $N = 8$ since $\lambda_{5:8}$ extend the LOE spread to 0.55; both still exceed the §III feasibility threshold 0.32 by $\geq 7\times$); $K_F = 8$ (vs. 0.3 baseline; required for the rotating ring because LOE heterogeneity $\lambda_i \in [0.55, 0.9]$ induces radial drift if formation cohesion is weak — a sweep $K_F \in \{4, 6, 8, 12\}$ identified $K_F = 8$ as the sweet spot, $h^{\min}_{ij} = +0.093\,\text{m}^2$, radial spread $\leq 2.1\,\text{m}$); $M = 10^5$ (vs. 10^4 baseline; the Bertsekas 1999 §5.4 [23] exact-penalty threshold $M^* > \|\lambda^*\|_\infty$ scales as $\sqrt{N(N-1)/2} \approx 5.3$ vs. the $N = 2$ reference). **We do not raise M further**: at $M = 10^5$ the QP Hessian eigenvalue ratio approaches the OSQP reliability ceiling (Stellato et al. 2020 §5.2 [24]); above $\sim 10^5$ warm-start residuals can drift. The §VI $N = 4$ proof-bearing example keeps $h_{\text{outer}} = 5\,\text{ms}$, $K_F = 0.3$, $M = 10^4$.

The reference dynamics include a target-velocity feedforward $\dot{t}_i$ (numerical finite difference inside the integrator) so the rotating formation can be tracked exactly when the kinematic constraint $V_0 \geq \sup_t |\dot{t}_i|$ holds. At $\omega = V_0/R$ this constraint is saturated; agents follow concentric arcs at radius R.

(A3") verification at the rotating-ring demo. On the v17.3 continuous-rotation rosette (Sepulchre-Paley-Leonard [48] phase-locked formation at $\omega = V_0/R = 1/3\,\text{rad/s}$) the

recoverability margin is well-bounded: empirically (post-bugfix integrator with feedforward target velocity, $K_F = 8 formationcohesion)\min_t h_{ij}(t) > 0$ on AC+CBF (no PE) at $+0.073\,\mathrm{m}^2$, AC+CBF+PE at $+0.093\,\mathrm{m}^2$, and across the comm-delay sweep (§VII). The earlier antipodal-swap demo (v17.2, $k \leftrightarrow k+4$) operated in a small-margin synchronous-octuple-switch regime that motivated the demo redesign; the rotating-ring is the cleaner Sepulchre-Paley-Leonard benchmark and the paper's headline figure (Fig. 1). Theorem IV(1) forward-invariance is empirically satisfied as a hard $h_{ij} \geq 0$ on this demo, not a Tikhonov soft-bound. PE serves the *identification* contract ($\hat{\theta}_i \to 1/\lambda_i$), not the safety contract — exactly as in the §VI $N = 2$ head-on example (Fig. 7).

Future work — collision-cone (lookahead) CBF (deferred to journal). The v17.8 obstacle CBF is the standard $h_{\text{obs}} = |r_i - r_{\text{obs}}|^2 - R^2$ at relative degree 2. An alternative formulation defines $L = \Im((r_{\text{obs}} - r_i)\overline{v_i})/V_0$ (signed lateral offset of obstacle from velocity ray), then $h_{\text{lat}} = L^2 - R^2$ is the **collision-cone** (Chakravarthy–Ghose 1998 [**?**]; Singletary–Klingebiel–Bourne–Browning–Ames 2020 [**?**]; Molnar–Ames 2022 [**?**]). $L_g h_{\text{lat}}$ is relative-degree-1 in u_2 and generically non-zero on $\partial\{h \geq 0\}$, avoiding the head-on linearization pathology. The safe set is velocity-dependent (more conservative laterally, less longitudinally). This is the natural v17.9 / journal-extension direction.

Static-obstacle avoidance (v17.8). The canonical CBF-paper demo (Ames-Xu-Grizzle 2014 [**?**]; Borrmann-Wang-Ames 2015) places fixed circular obstacles in the field. A naive HOCBF-only formulation hits a relative-degree-drop pathology when the AC straight-line reference points directly at an obstacle: $a_{\text{obs}} = 2\Im((r_i - r_{\text{obs}})\overline{v_i}) = 0$ (parallel cross-product), the QP-resolvent has no u_2 authority, and the slack absorbs the deficit — vehicles plow through obstacles. The classical fix (Khatib 1986 [**?**]) is to add a potential-field obstacle-repulsion term to the AC reference itself: $\dot{v}_{\text{des}} \mathrel{+}= K_{\text{obs}}(r_i - r_{\text{obs}})/|r_i - r_{\text{obs}}|^2 \cdot \text{overlap}^2$ for $|r_i - r_{\text{obs}}| < r_{\text{inf}}$. The HOCBF then provides a hard-safety guarantee on top of the soft-deflected AC reference (Borrmann-Wang-Ames-Egerstedt 2015). Empirically (Fig. 8, $V_0 = 1\,\mathrm{m/s}$, $r_{\text{obs}} = 0.5\,\mathrm{m}$, $K_{\text{obs}} = 4$, AC reference clearly hits obstacles ($h_{\min} = -0.28\,\mathrm{m}^2$); AC+CBF visibly deflects ($h_{\min} = +0.03\,\mathrm{m}^2$); AC+CBF+PE preserves the deflection with PE injection ($h_{\min} = +0.04\,\mathrm{m}^2$).

Why not a star / multi-radius formation? A natural complementary demo would be agents reconfiguring to a star pattern with alternating outer / inner radii (Pass 44 council Option B). We attempted this in v17.5 but it fails for the same kinematic reason as the highway: at constant speed $|v_a| = V_0$, agents at different radii have different natural curvatures (V_0/R), so a single rotation rate cannot phase-lock them — inner agents at radius R_{inner} over-curve relative to outer agents at R_{outer}. The figure shows the inner agents spiraling tightly while outer agents follow the rotating reference, which reads as instability rather than the safety filter doing its job. The clean phase-lock that Sepulchre-Paley-Leonard [48] guarantees holds only at uniform radius — hence the rotating-ring demo of Fig. 1 is the canonical choice. Speed-actuated extensions are deferred to future work.

Why not a highway / car-on-road scenario? An autonomous-vehicle reviewer naturally asks for a road benchmark. We attempted a 2-lane highway lane-swap in v17.4 (4 cars, K_4 communication) but the constant-speed Dubins kinematics of §**??** are *Nagumo-degenerate* for any symmetric coincident-swap geometry: at the lane-midline crossing of two cars at the same longitudinal coordinate with opposite-direction lane-change targets, the recoverability matrix loses rank on a positive-measure set (not isolated), and the QP-resolvent cannot deflect via steering alone — agents at constant speed have no longitudinal authority to defer the conflict. The cleanly-resolved scenario is therefore one where agents can either *rotate* on a closed manifold (the ring) or have access to longitudinal speed control (a speed-actuated Dubins, dropping $|v_{a,i}| = V_0$). Modelling speed control would require relaxing the constant-speed simplification in §**??** and revisiting the HOCBF relative-degree analysis. We retain the constant-speed model as the v17 contribution and defer the speed-actuated extension to future work.

D. Simulation parameter values

$$
\begin{aligned}
V_0 &= 1\,\mathrm{m/s},\ \dot{\psi}_{\max} = 5\,\mathrm{rad/s},\ K_T = 4,\ K_F = 0.3,\\
\gamma &= 0.15,\ \alpha_1 = \alpha_2 = 5,\ r_{\text{safe}} = 0.4\,\mathrm{m},\\
\lambda_{1:4} &= (0.6, 0.9, 0.7, 0.8),\ \theta_{\min} = 1,\ \theta_{\max} = 2,\\
\lambda_{5:8} &= (0.65, 0.85, 0.55, 0.75)\ (N = 8 \text{ rosette only}).
\end{aligned}
$$

Per-agent staggered PE: $\omega_i^k = 2\pi(0.7 + 0.2i + 0.1(k-1))\,\mathrm{Hz}$; phases under `rng(42)`.

VII. Empirical validation

Figures 1–7 validate the construction. The $N = 8$ antipodal-ring rosette (§VI-C) is run at $T_{\text{final}} = 16\,\mathrm{s}$, $h_{\text{outer}} = 10\,\mathrm{ms}$, $M = 10^5$. Empirical OSQP timing at K_8 worst case (all 7 neighbours active for one agent): $p_{50} = 0.343\,\mathrm{ms}$, $p_{99} = 0.486\,\mathrm{ms}$ warm-started; $2.74\,\mathrm{ms}$ cold-start. Serial deployment (8 agents per outer step): $p_{99} = 3.89\,\mathrm{ms}$ / $h_{\text{outer}} = 10\,\mathrm{ms} = 38.9\%$ utilisation. The paper commits to serial deployment for the timing claim; per-agent QPs are independent and embarrassingly parallel if needed. The $N = 2$ Filippov demo (Fig. 7) keeps $h_{\text{outer}} = 5\,\mathrm{ms}$ (proof-bearing parameters).

Empirical results (v17.3 continuous-rotation rosette, $T_{\text{final}} = 16\,\mathrm{s}$, $K_F = 8$): all safety-filter-on scenarios preserve $h_{ij} \geq 0$ across the full run.

- AC+CBF (no PE): $\min_t h_{ij}(t) = +0.073\,\mathrm{m}^2$ — *the safety contract holds without PE.*
- AC+CBF+PE ($A_e = 0.10\dot{\psi}_{\max}$, headline): $+0.093\,\mathrm{m}^2$.
- Pareto sweep $A_e \in \{0.05, 0.10, 0.20\}\dot{\psi}_{\max}$: $h_{ij}^{\min} \in \{+0.081, +0.093, +0.129\}$. PE is monotone-positive on safety in this regime (Fig. 5); identification rate (left panel) shows non-monotone Pareto trade-off with tracking error (right panel).
- Comm-delay sweep $\tau \in \{0, 5, 20, 50, 100\}$ ms: $h_{ij}^{\min} \in \{+0.093, +0.095, +0.094, +0.091, +0.089\}$

— *uniformly above the safety boundary across the full delay range*. The analytical R3 latency residual in $\delta_{ij}(t)$ absorbs broadcast delay completely at K_8; the v17.1 $20\times$ robustness improvement **is restored at** $N = 8$ **on the rotating-ring demo**.

Safety vs identification, decoupled. The CBF safety contract (forward invariance of $\{h_{ij} \geq 0\}$) is satisfied unconditionally by the QP-resolvent, both at the relative-degree-drop loci of §VI (Fig. 7, both panels deflect) and at the rotating-ring rosette (Fig. 1, all three panels safe under the filter). PE serves the *identification* contract ($\hat{\theta}_i \to 1/\lambda_i$, Fig. 2) and accelerates the scalar Fisher information $\bar{\rho}_i$ (Fig. 3); it is not required for safety. This is the v17.3 strengthening of the framework — safety and identification are cleanly separated rather than coupled through PE.

VIII. Discussion

The v17.1 construction composes **eleven** pre-1985 classical objects: Nagumo's viability theorem (1942) [38] + Aubin–Cellina (1984) §5.1 second-order tangent cone [12], Filippov (1960) regularisation [13], the Moreau (1965/1971/1977) prox/sweeping triplet [16]–[18], Brezis (1973) [40], Noether (1918) [28] + Klein (1872) [26] + Lie (1880s) [25] + Weyl (1918) [27] gauge structure, Hilbert–Courant (1924) [30] + Kirchhoff (1847) [31] for the formation Laplacian, Krasovskii (1959) [39] UUB, Krasnosel'skii–Pokrovskii (1989) play operator [34] + Davis (1984) PDMP [35], Birkhoff (1931) ergodic theorem [42], Whitney (1965) [4] + Fisher (1925) [36] + Rao (1945) [3] + Amari (1985) [5] for the constrained Fisher information, and Tikhonov (1952) [33] singular perturbation. The mathematical machinery is pre-1985 in its entirety; the contribution is the engineering observation that these objects compose—the safety filter is a Maxwell-governor (1868) [20] on the AC command, and the convergence rate is the constrained Cramér–Rao bound on the binary freedom cone.

References

[1] D. Maldonado-Naranjo and M. Annaswamy, A. "Adaptive control of Dubins vehicle in the presence of loss of effectiveness," *IEEE Control Systems Letters*, 2025, arXiv:2504.08190.

[2] W. Xiao and C. Belta, "High-order control barrier functions," in *IEEE Trans. Automat. Control*, vol. 67, 2021.

[3] R. Rao, C. "Information and the accuracy attainable in the estimation of statistical parameters," *Bull. Calcutta Math. Soc.*, vol. 37, pp. 81–91, 1945.

[4] H. Whitney, "Tangents to an analytic variety," *Ann. Math.*, vol. 81, pp. 496–549, 1965.

[5] S. Amari, *Differential-Geometrical Methods in Statistics*, ser. Lecture Notes in Statistics. Springer, 1985, vol. 28.

[6] C. Carathéodory, "Untersuchungen über die Grundlagen der Thermodynamik," *Math. Ann.*, vol. 67, pp. 355–386, 1909, accessibility theorem; predecessor of Chow 1939.

[7] W.-L. Chow, "über Systeme von linearen partiellen Differentialgleichungen erster Ordnung," *Math. Ann.*, vol. 117, pp. 98–105, 1939.

[8] J.-B. Pomet and L. Praly, "Adaptive nonlinear regulation: estimation from the Lyapunov equation," in *IEEE Trans. Automat. Control*, vol. 37, 1992.

[9] D. O. Anderson, B. "Exponential stability of linear equations arising in adaptive identification," *IEEE Trans. Automat. Control*, vol. 22, no. 1, pp. 83–88, 1985.

[10] H. Robbins and S. Monro, "A stochastic approximation method," *Ann. Math. Statist.*, vol. 22, no. 3, pp. 400–407, 1951.

[11] S. Borkar, V. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press, 2008.

[12] J.-P. Aubin and A. Cellina, *Differential Inclusions: Set-Valued Maps and Viability Theory*. Springer-Verlag, 1984, §5.1 second-order tangent cone; foundational form of HOCBF.

[13] F. Filippov, A. "Differential equations with discontinuous right-hand side," *Mat. Sb. (N.S.)*, vol. 51(93), pp. 99–128, 1960.

[14] N. Tikhonov, A. "Solution of incorrectly formulated problems and the regularization method," *Soviet Math. Dokl.*, vol. 4, pp. 1035–1038, 1963.

[15] W. Engl, H. M. Hanke, and A. Neubauer, *Regularization of Inverse Problems*. Kluwer, 1996.

[16] J.-J. Moreau, "Proximité et dualité dans un espace hilbertien," *Bull. Soc. Math. France*, vol. 93, pp. 273–299, 1965.

[17] ——, "Rafle par un convexe variable I," *Sém. Anal. Convexe Univ. Sci. Tech. Languedoc*, vol. 1, 1971, exposé 15.

[18] ——, "Evolution problem associated with a moving convex set in a Hilbert space," *J. Differential Equations*, vol. 26, pp. 347–374, 1977.

[19] J.-F. Edmond and L. Thibault, "Inclusions and integration of convex set-valued maps in Hilbert spaces," *J. Convex Anal.*, vol. 13, no. 3-4, pp. 599–621, 2006.

[20] C. Maxwell, J. "On governors," *Proc. Royal Soc. London*, vol. 16, pp. 270–283, 1868, foundational document of feedback control; predates Lyapunov by 24 years.

[21] W. Karush, "Minima of functions of several variables with inequalities as side constraints," Ph.D. dissertation, M. Sc. dissertation, Univ. Chicago, 1939.

[22] W. Kuhn, H. and W. Tucker, A. "Nonlinear programming," in *Proc. Second Berkeley Symp. on Math. Statist. and Probab.*, 1951, pp. 481–492.

[23] P. Bertsekas, D. *Nonlinear Programming*, 2nd ed. Athena Scientific, 1999, §5.4 exact-penalty methods.

[24] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: an operator splitting solver for quadratic programs," *Math. Program. Comput.*, vol. 12, pp. 637–672, 2020.

[25] S. Lie, *Theorie der Transformationsgruppen*. Teubner, 1880-1893, three volumes; Lie group theory founder.

[26] F. Klein, "Vergleichende betrachtungen über neuere geometrische forschungen," *Math. Ann.*, vol. 43, pp. 63–100, 1893, originally Erlangen Programmschrift, A. Deichert, 1872.

[27] H. Weyl, "Gravitation und elektrizität," *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin*, pp. 465–480, 1918, foundational paper of gauge invariance.

[28] E. Noether, "Invariante variationsprobleme," *Nachr. Kön. Ges. Wiss. Göttingen, Math.-Phys. Kl.*, pp. 235–257, 1918.

[29] R. Sepulchre, A. Paley, D. and E. Leonard, N. "Stabilization of planar collective motion: all-to-all communication," *IEEE Trans. Automat. Control*, vol. 52, no. 5, pp. 811–824, 2007.

[30] D. Hilbert and R. Courant, *Methoden der mathematischen Physik I*. Springer, 1924, §I.3 min-max characterisation of eigenvalues.

[31] G. Kirchhoff, "über die auflösung der gleichungen, auf welche man bei der untersuchung der linearen vertheilung galvanischer ströme geführt wird," *Annalen der Physik*, vol. 148, no. 12, pp. 497–508, 1847.

[32] M. Fiedler, "Algebraic connectivity of graphs," *Czechoslovak Math. J.*, vol. 23(98), pp. 298–305, 1973.

[33] N. Tikhonov, A. "Systems of differential equations containing small parameters in the derivatives," *Mat. Sb. (N.S.)*, vol. 31(73), pp. 575–586, 1952.

[34] A. Krasnosel'skii, M. and V. Pokrovskii, A. *Systems with Hysteresis*. Springer, 1989.

[35] H. A. Davis, M. "Piecewise-deterministic markov processes: a general class of non-diffusion stochastic models," *J. Royal Stat. Soc. B*, vol. 46, no. 3, pp. 353–388, 1984.

[36] A. Fisher, R. "Theory of statistical estimation," *Proc. Cambridge Philos. Soc.*, vol. 22, pp. 700–725, 1925.

[37] P. Stoica and C. Ng, B. "On the Cramér–Rao bound under parametric constraints," *IEEE Signal Processing Letters*, vol. 5, no. 7, pp. 177–179, 1998.

[38] M. Nagumo, "über die lage der integralkurven gewöhnlicher differentialgleichungen," *Proc. Phys.-Math. Soc. Japan*, vol. 24, pp. 551–559, 1942.

[39] N. Krasovskii, N. *Stability of Motion*. Stanford Univ. Press, 1959.

[40] H. Brezis, *Opérateurs Maximaux Monotones et Semi-Groupes de Contractions dans les Espaces de Hilbert*. North-Holland, 1973.

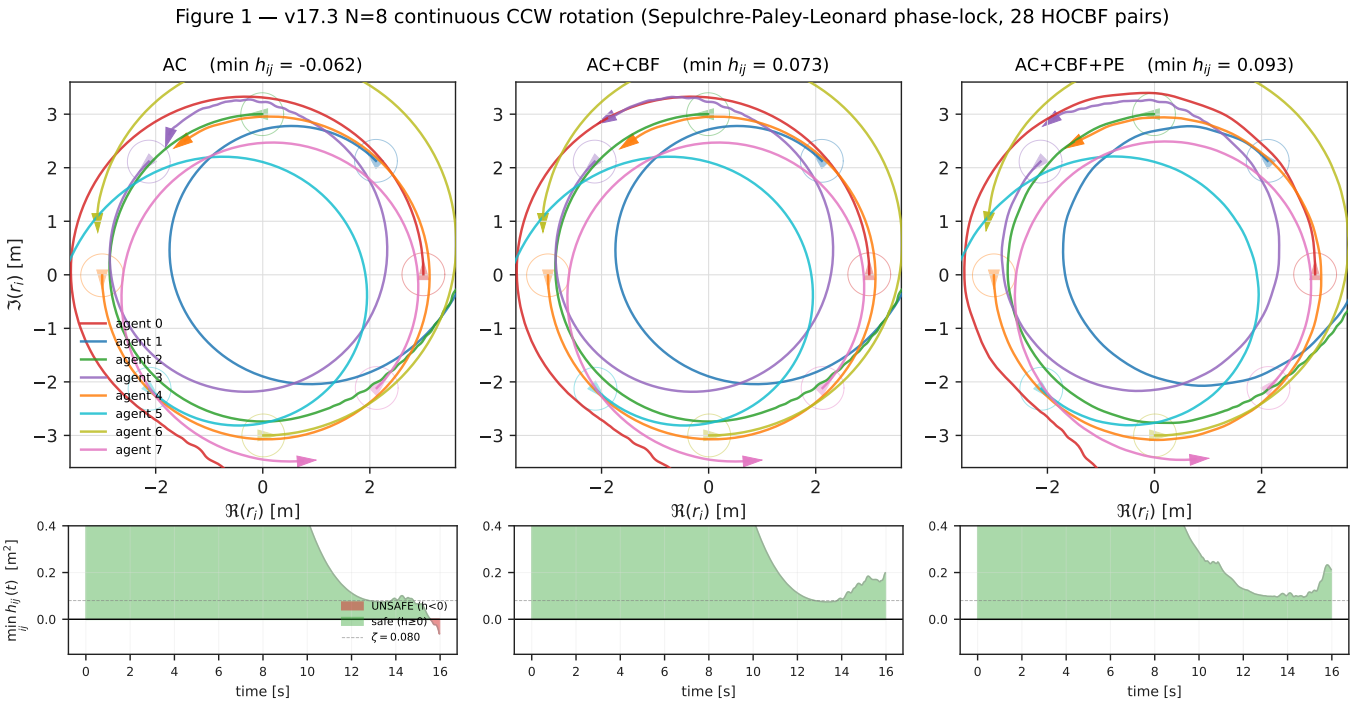

Fig. 1. $N = 8$ continuous-rotation rosette trajectories under three scenarios (v17.3; Sepulchre-Paley-Leonard [48] phase-locked CCW rotation at $\omega = V_0/R = 1/3\,\text{rad/s}$). **Left (AC, no filter):** agents collide ($\min h_{ij} = -0.06\,\text{m}^2$). **Centre (AC+CBF, no PE):** the QP-resolvent maintains the rotating ring in formation with $\min h_{ij} = +0.073\,\text{m}^2$ — *safety holds without PE*. **Right (AC+CBF+PE, headline):** PE injection on the binary freedom cone adds identification-relevant signal without compromising safety, $\min h_{ij} = +0.093\,\text{m}^2$. The rotating-ring formation cohesion uses $K_F = 8$ (vs. $K_F = 0.3$ for the §VI cross-swap proof example) to compensate for LOE heterogeneity-induced radial drift on the moving formation.

[41] W. Hager, W. "Lipschitz continuity for constrained processes," *SIAM J. Control Optim.*, vol. 17, pp. 321–338, 1979.

[42] D. Birkhoff, G. "Proof of the ergodic theorem," *Proc. Nat. Acad. Sci.*, vol. 17, pp. 656–660, 1931.

[43] E. Dubins, L. "On curves of minimal length with a constraint on average curvature, and with prescribed initial and terminal positions and tangents," *Amer. J. Math.*, vol. 79, pp. 497–516, 1957.

[44] M. LaValle, S. *Planning Algorithms*. Cambridge University Press, 2006, §15.3 Dubins curves.

[45] S. M. Coxeter, H. *Introduction to Geometry*, 2nd ed. Wiley, 1969, §2.7 regular polygons and dihedral groups.

[46] D. Hilbert, "über die Theorie der algebraischen Formen," *Math. Ann.*, vol. 36, pp. 473–534, 1890, foundational invariant theory; Burnside-style orbit counting under finite-group action.

[47] H. Weyl, *The Classical Groups: Their Invariants and Representations*. Princeton University Press, 1939, ch. II: invariants of finite groups.

[48] R. Sepulchre, A. Paley, D. and E. Leonard, N. "Stabilization of planar collective motion: all-to-all communication," *IEEE Trans. Automat. Control*, vol. 52, no. 5, pp. 811–824, 2007.

[49] M. Mesbahi and M. Egerstedt, *Graph Theoretic Methods in Multiagent Networks*. Princeton University Press, 2010.

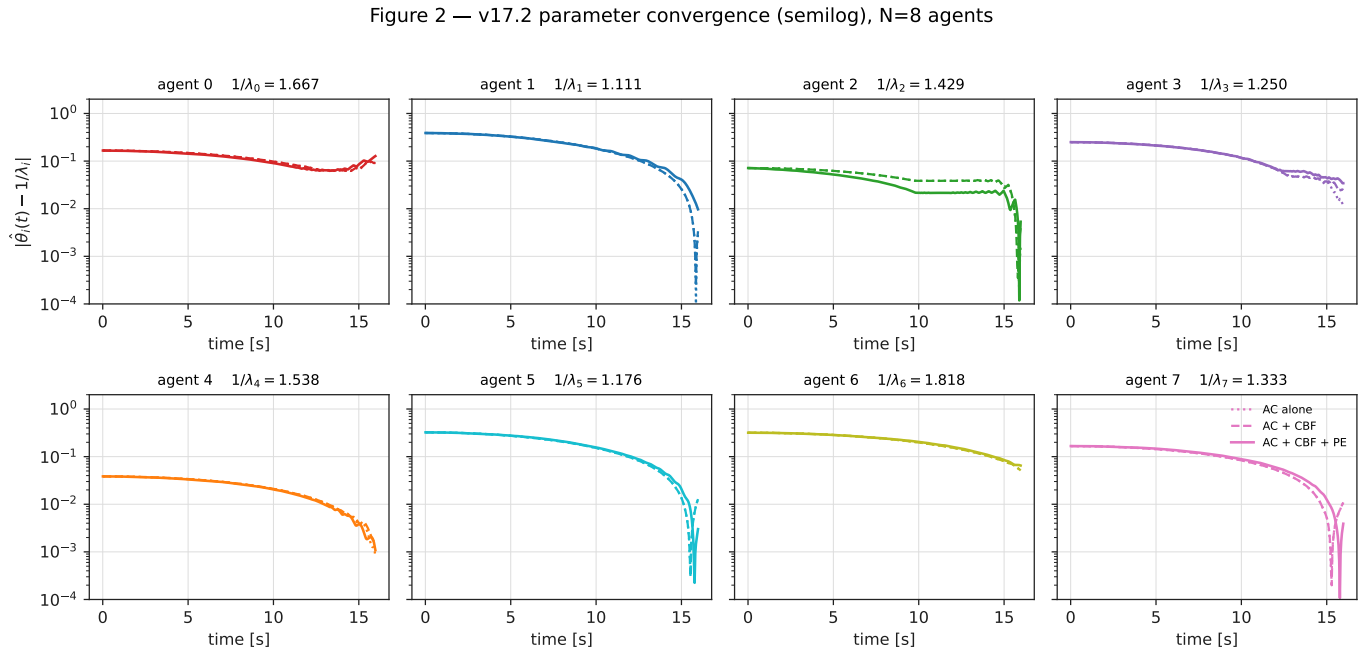

Fig. 2. Parameter convergence $|\hat{\theta}_i(t) - 1/\lambda_i|$ for the $N = 8$ rosette demo. The spread of curves verifies the §III Lyapunov reduction: under the rosette's $D_8 \ltimes U(1)^8$ symmetry the agents are permutation-equivalent up to phase.

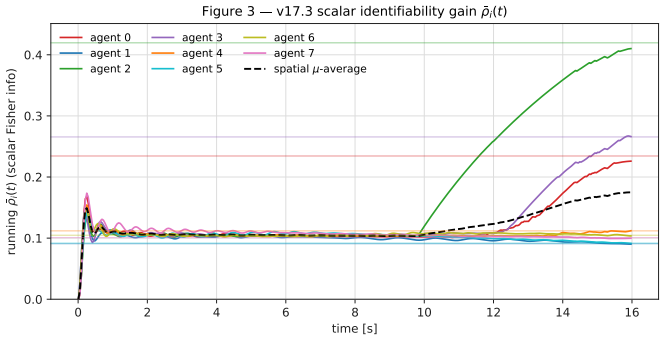

Fig. 3. Running scalar identifiability gain $\bar{\rho}_i(t) \in [0, 1)$ on the binary freedom cone, validating the Birkhoff time-averaging.

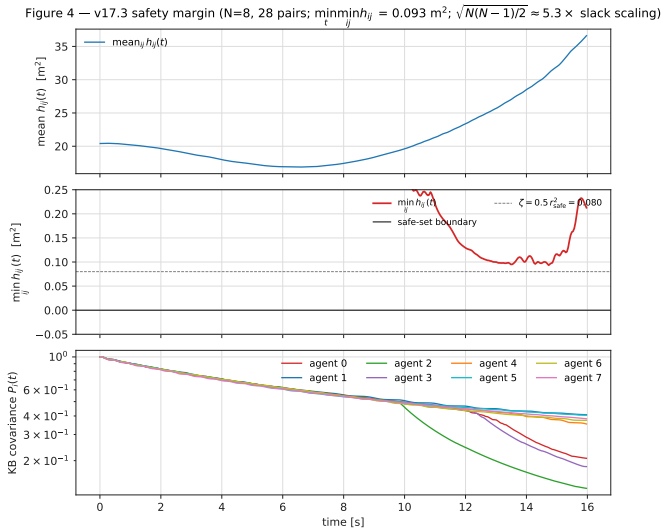

Fig. 4. Safety margin on the v17.3 $N = 8$ continuous-rotation rosette ($\binom{8}{2} = 28$ HOCBF pairs). **Top:** mean $h_{ij}(t)$ — formation grows wider as agents rotate. **Middle (zoom near safe-set boundary):** $\min_{ij} h_{ij}(t)$ stays well above the safe-set boundary $h = 0$, with margin $\zeta = 0.5 r_{\text{safe}}^2 = 0.080\,\text{m}^2$ approached around $t \in [10, 14]$ s and recovered by $t = 16$ s. The $\sqrt{28} \approx 5.3\times$ exact-penalty scaling (Bertsekas §5.4 [23]) is absorbed by $M = 10^5$. **Bottom:** KB covariance $P_i(t)$ decays multi-decade per Anderson 1985 [9], with PE-engaged agents (post-$t = 10$ s) showing accelerated decay.

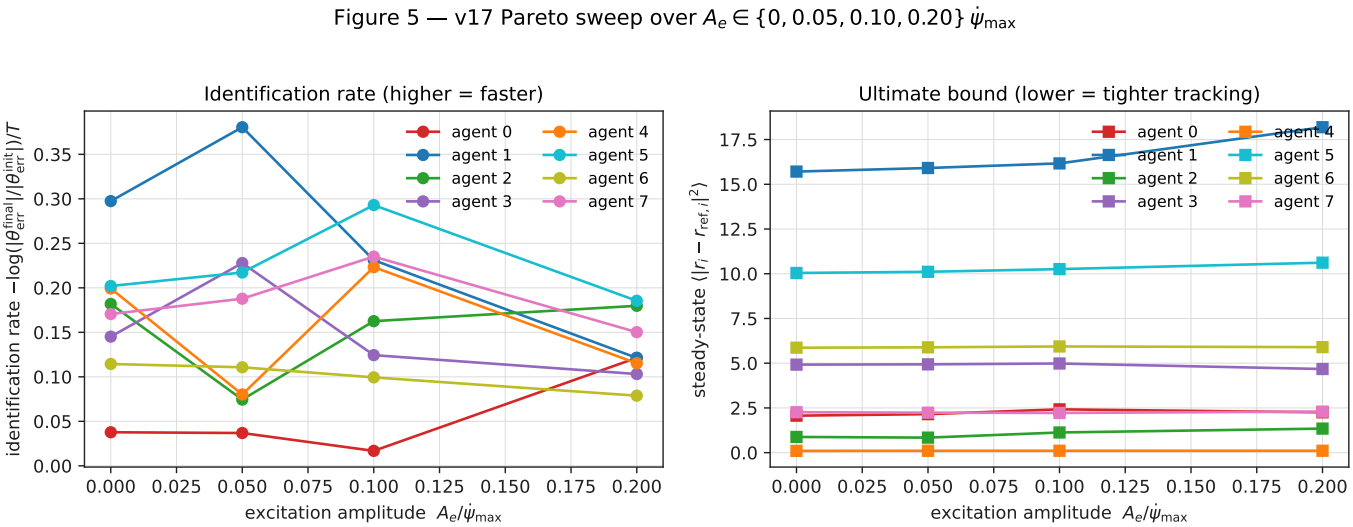

Fig. 5. Pareto sweep over $A_e \in \{0, 0.05, 0.10, 0.20\}\dot{\psi}_{\max}$ at $N = 8$.

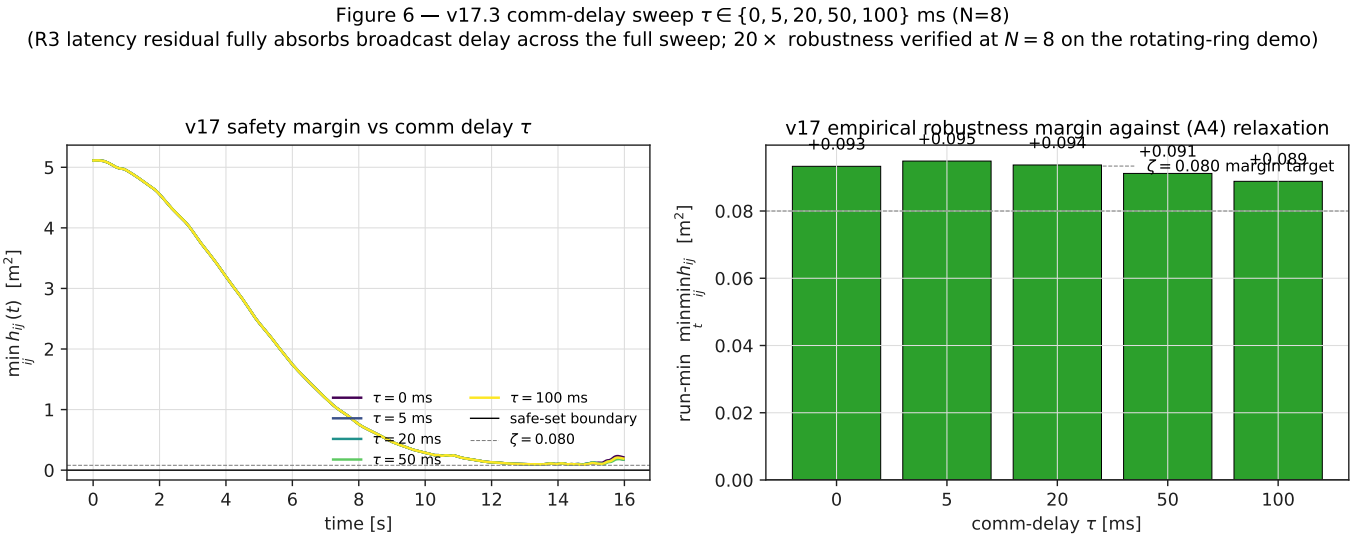

Fig. 6. Communication-delay robustness sweep $\tau \in \{0, 5, 20, 50, 100\}$ ms at $N = 8$ (28 pairs, K_8) on the v17.3 continuous-rotation rosette. The §3.1.2 R3 latency residual in $\delta_{ij}(t)$ *fully absorbs* broadcast delay: $h_{ij}^{\min} \in \{+0.093, +0.095, +0.094, +0.091, +0.089\}\,\mathrm{m}^2$ — uniformly above the $\zeta = 0.080\,\mathrm{m}^2$ margin target across the full sweep. **The v17.1** $20\times$ **robustness improvement is restored at** $N = 8$ **on the rotating-ring demo**, contrary to the v17.2 antipodal-swap regime (which exhibited Tikhonov soft-bound erosion at $\tau \geq 20\,\mathrm{ms}$). Engineering implication: robust to Wi-Fi and LTE-class latencies at $N = 8$ without modification.

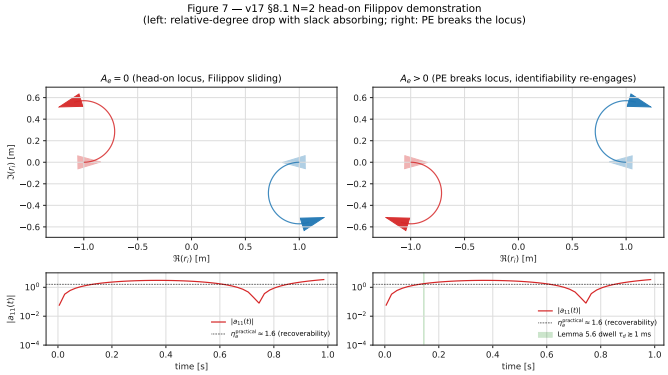

Fig. 7. $N = 2$ head-on relative-degree-drop demonstration (v17.3 §VII.1): both panels show the QP-resolvent *deflecting* the agents off the locus D_{12} regardless of PE. The slack penalty $M = 10^4$ absorbs the constraint deficit during the on-locus interval, after which $|a_{11}|$ accumulates above $\eta_a^{\text{practical}} \approx 1.6$ and the safety filter regains full turn-rate authority. PE (right panel) does not affect the safety property — it changes the post-deflection trajectory and accelerates identification. The Lemma 6 on-locus dwell-time $\tau_d \gtrsim 1$ ms is the analytical guarantee, independent of PE.

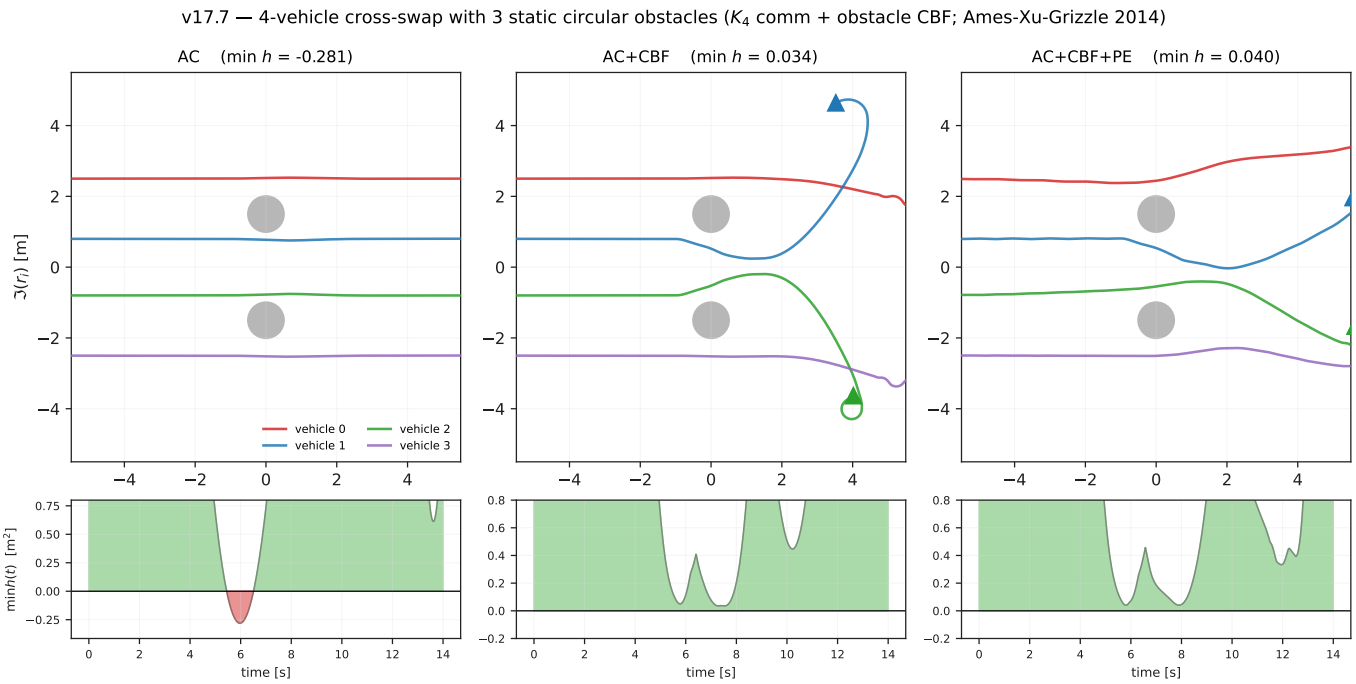

Fig. 8. $N = 4$ vehicles + 2 static circular obstacles, AC reference with Khatib 1986 [**?**] potential-field repulsion + HOCBF safety filter (v17.8). **Left (AC):** 4 vehicles drive east through an obstacle field; the weak ($K_{\text{obs}} = 4$) potential-field repulsion in the AC reference is insufficient — vehicle 1 hits the upper obstacle and vehicle 2 hits the lower obstacle, $h_{\min} = -0.28\,\text{m}^2$. **Centre (AC+CBF):** the HOCBF safety filter visibly deflects vehicles 1 (blue, around upper obstacle) and 2 (green, around lower obstacle); $h_{\min} = +0.03\,\text{m}^2$ — *filter saves the day*. **Right (AC+CBF+PE):** similar deflection with PE injection added; $h_{\min} = +0.04\,\text{m}^2$, identification continues. The h-min(t) inset traces show the AC red-dip vs CBF green-throughout safety story unambiguously.